

BIG VALLEY AGCM, AB, Canada

WMO: 710110

Lat: 51.9978N

Lon: 112.8025W

Elev: 2854

StdP: 13.24

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-20.9	-15.7	-27.6	1.3	-20.4	-22.0	1.8	-15.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	83.6	61.7	80.1	60.2	76.9	59.4	65.0	77.2	63.0	75.4	61.3	73.5	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.6	88.0	70.5	58.3	80.8	68.9	56.3	75.2	66.7	31.6	77.0	30.0	76.0	28.7	73.9	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-27.1	89.8	4.7	4.4	-30.5	93.0	-33.2	95.5	-35.9	98.0	-39.2	101.3
			WB	-27.4	69.2	4.6	2.4	-30.7	71.0	-33.4	72.4	-36.0	73.8	-39.4	75.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.4	15.5	13.9	25.5	38.7	51.4	58.1	62.7	61.6	53.0	40.4	25.3	13.3	
	DBStd	21.11	14.85	14.33	13.69	9.32	8.19	5.66	5.26	5.97	8.48	9.07	12.92	13.63	
	HDD50	5535	1068	1010	759	350	81	6	0	1	63	315	743	1139	
	HDD65	9789	1533	1430	1224	788	424	218	108	138	369	761	1192	1604	
	CDD50	1312	0	0	0	12	126	248	394	360	152	19	1	0	
	CDD65	90	0	0	0	0	4	10	37	32	7	0	0	0	
	CDH74	1374	0	0	0	8	95	135	484	473	172	7	0	0	
	CDH80	311	0	0	0	0	9	19	119	122	42	0	0	0	
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Precipitation	PrecAvg	15.1	0.6	0.4	0.6	1.0	1.9	3.0	2.6	2.1	1.3	0.6	0.6	0.5	
	PrecMax	19.3	1.3	0.9	2.1	2.4	3.7	6.0	5.9	4.3	3.2	1.4	1.9	1.1	
	PrecMin	9.0	0.0	0.0	0.1	0.1	0.2	1.3	0.9	0.1	0.2	0.2	0.1	0.0	
	PrecStd	2.6	0.3	0.2	0.4	0.6	0.9	1.2	1.2	1.1	0.7	0.4	0.4	0.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.9	45.4	57.6	71.7	81.0	82.5	87.8	88.2	85.3	72.8	59.0	45.3	
		MCWB	37.4	37.6	43.3	49.1	55.0	59.7	64.3	62.2	58.2	52.4	45.8	37.3	
	2%	DB	42.5	39.9	52.0	66.0	76.5	77.8	83.5	83.1	79.0	64.5	50.4	40.3	
		MCWB	35.8	33.8	40.7	47.3	53.3	59.3	63.6	61.6	56.5	49.4	41.0	34.1	
	5%	DB	38.8	36.8	47.0	60.0	72.0	74.3	79.2	79.4	74.4	59.7	44.9	35.9	
		MCWB	33.4	31.8	37.9	43.6	52.1	57.8	62.5	61.0	55.6	46.6	37.4	30.7	
	10%	DB	35.1	33.5	42.2	54.5	67.3	71.0	75.6	75.4	69.0	54.7	40.7	31.7	
		MCWB	30.7	29.7	35.4	41.1	50.0	56.1	61.6	60.3	53.5	43.8	34.5	27.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.5	37.7	45.2	50.7	56.8	64.5	69.1	66.7	60.5	53.9	46.4	38.2
			MCDB	44.8	44.5	56.4	70.8	75.1	76.1	80.7	77.8	79.1	70.1	57.8	43.7
2%		WB	36.1	34.3	41.2	47.3	54.8	61.2	65.8	64.5	58.3	50.5	41.4	34.2	
		MCDB	41.8	39.0	50.2	64.6	72.2	74.0	77.8	77.6	75.0	61.9	49.5	39.8	
5%		WB	33.6	32.1	38.3	44.5	53.2	59.2	63.7	62.7	56.2	47.4	38.1	31.0	
		MCDB	38.5	36.1	46.4	57.9	69.0	70.8	75.6	76.2	71.1	58.5	44.1	35.7	
10%		WB	30.8	29.8	35.4	41.9	51.3	57.5	62.1	60.9	54.3	44.6	35.1	27.9	
		MCDB	35.0	33.6	41.6	52.9	65.0	68.0	73.6	73.0	68.3	54.0	39.4	31.4	
Mean Daily Temperature Range	5% DB	MDBR	16.2	17.1	17.5	20.1	21.6	20.2	21.6	22.5	22.5	20.2	16.6	15.1	
		MCDBR	17.9	19.4	23.3	30.7	29.0	25.3	27.7	29.5	32.0	28.5	22.5	18.4	
		MCWBR	13.2	14.5	13.8	15.3	12.4	11.3	12.2	12.3	14.0	15.2	14.3	13.8	
	5% WB	MCDBR	17.9	19.3	22.5	28.7	25.7	22.5	24.1	25.3	28.7	27.1	21.3	18.2	
		MCWBR	13.5	14.8	14.0	14.5	11.2	11.3	12.5	11.9	13.1	14.7	14.1	13.9	
Clear-Sky Solar Irradiance	taub	0.265	0.269	0.285	0.330	0.350	0.354	0.352	0.366	0.315	0.281	0.253	0.243		
	taud	2.312	2.401	2.451	2.354	2.348	2.374	2.407	2.349	2.462	2.538	2.480	2.352		
	Ebn at Noon	238	276	293	288	286	285	284	272	277	269	248	230		
	Edh at Noon	19	24	28	35	37	36	35	35	28	21	16	16		
All-Sky Solar Radiation	RadAvg	345	675	1116	1501	1823	1939	2050	1667	1165	699	364	262		
	RadStd	29	56	89	130	117	120	104	116	124	82	33	26		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (84 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page